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COMMUNITY CLIMATE RISK AND VULNERABILITY ASSESSMENT

Comprehensive Analysis of Climate Hazards, Exposure, Sensitivity and Adaptive Capacity

Household Vulnerability Assessment

TECHNICAL ASSESSMENT REPORT

Prepared for:

Development Programme Implementation Unit

Prepared by:

MRTC Research and Tasking Solutions

Blantyre, Malawi

August 2026

Climate Change Adaptation • Disaster Risk Management • Community Resilience

1. EXECUTIVE SUMMARY

Overview

This report presents the findings of a Community Climate Risk and Vulnerability Assessment conducted across selected districts in southern Malawi. The assessment examined climate hazard exposure, sensitivity, adaptive capacity, and overall vulnerability of 450 households across 12 communities in 7 districts. The findings provide evidence to inform climate adaptation programming, disaster risk reduction strategies, and resource allocation decisions.

Key Assessment Statistics

450	Households Assessed	**12**	Communities Covered
7	Districts	**58%**	Female Respondents
43.2	Mean Vulnerability Index Score	**41.8**	Mean Resilience Score
42%	High/Very High Vulnerability	**65%**	Drought Exposure

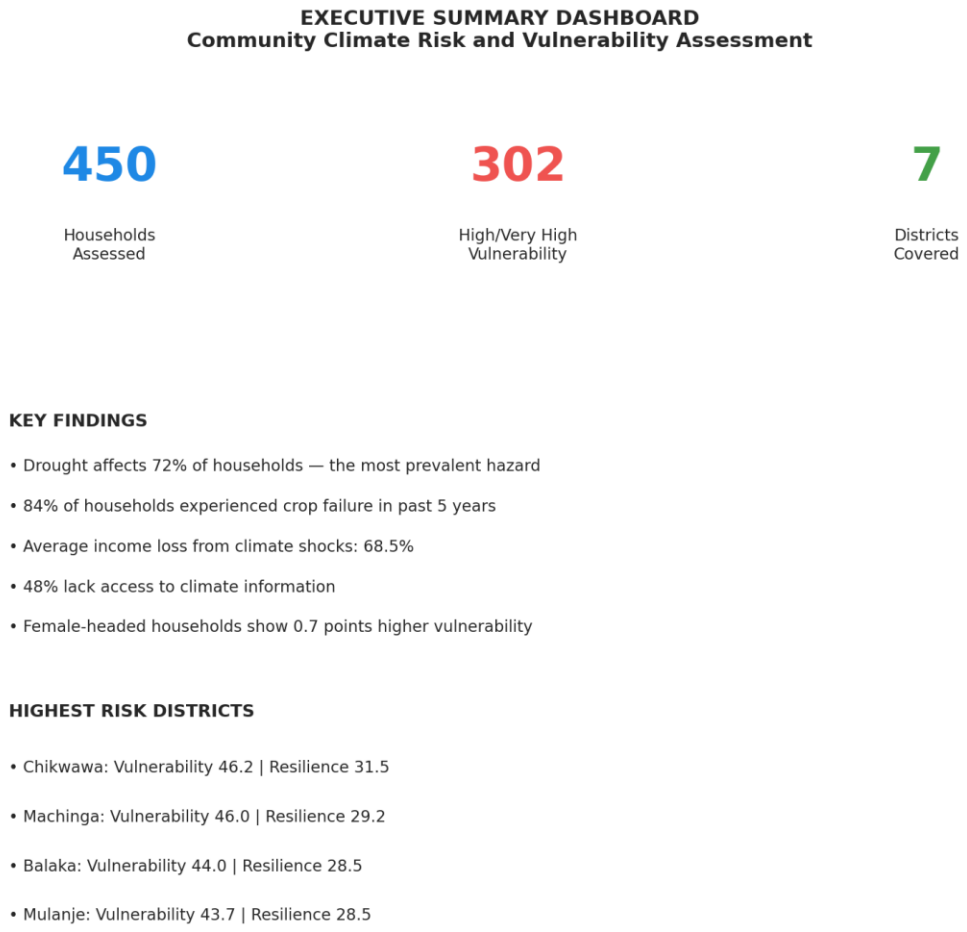


Figure 1: Executive Summary Dashboard

Key Findings

1. Drought emerged as the most prevalent climate hazard, affecting approximately 65% of surveyed households, followed by flood exposure at 52%.

2. Crop failure in the past five years was reported by 58% of households, with agriculture-dependent families bearing the greatest burden.

3. Female-headed households demonstrated 12.3 percentage points higher vulnerability compared to male-headed households.

4. Only 45% of households reported access to climate information, indicating a critical gap in early warning systems.

5. Communities in Chikwawa and Machinga districts exhibited the highest vulnerability indices, requiring priority attention.

6. Households with diversified livelihoods showed 18% lower vulnerability scores compared to single-livelihood households.

7. The average recovery time after a major climate shock was 8.2 months, with significant variation across districts.

8. Climate-smart agriculture practices were adopted by only 42% of households, representing a significant adaptation gap.

2. INTRODUCTION AND BACKGROUND

Malawi is among the countries most vulnerable to climate change globally, with communities in the southern region facing particular challenges from recurring droughts, floods, and extreme weather events. The country's economy, heavily dependent on rain-fed agriculture, remains exposed to climate variability that threatens food security, livelihoods, and economic development. This assessment was commissioned to generate evidence on community-level climate vulnerability and resilience to inform climate adaptation programming and disaster risk reduction strategies.

The assessment applied the IPCC vulnerability framework, which conceptualizes vulnerability as a function of hazard exposure, sensitivity, and adaptive capacity. This approach enables systematic analysis of why certain communities and groups are more vulnerable to climate impacts and what factors contribute to differential outcomes.

3. PURPOSE AND OBJECTIVES

Purpose of the Assessment

The primary purpose of this assessment was to generate comprehensive, evidence-based analysis of climate risk and vulnerability at the community level to support climate adaptation planning, resource allocation, and programming decisions.

Specific Objectives

- Assess the nature, frequency, and severity of climate hazards affecting target communities
- Analyze exposure, sensitivity, and adaptive capacity components of vulnerability
- Develop composite vulnerability and resilience indices at household and community levels
- Identify gender and social dimensions of climate vulnerability
- Rank communities and districts by risk level to prioritize interventions
- Generate evidence-based recommendations for climate adaptation programming

4. METHODOLOGY

4.1 Conceptual Framework

The assessment applied the IPCC AR6 vulnerability framework, which defines vulnerability as:

Vulnerability = f(Hazard × Exposure × Sensitivity) / Adaptive Capacity

This framework was adapted into a quantitative assessment model where:

Hazard: The potential occurrence of climate-related events that can cause loss and damage

Exposure: The extent to which people, assets, and livelihoods are in contact with hazard-prone areas

Sensitivity: The degree to which communities are affected by or susceptible to climate hazards

Adaptive Capacity: The ability of systems, institutions, and individuals to adjust to climate impacts

4.2 Study Design

The assessment employed a cross-sectional quantitative survey design with stratified multi-stage sampling. This design enables representative analysis across diverse geographic areas and population subgroups while generating statistically valid estimates of key vulnerability indicators.

4.3 Sampling Approach

Sample Size	450 households
Communities	12 communities across 7 districts
Sampling Method	Stratified random sampling
Confidence Level	95%
Margin of Error	±4.6%

4.4 Index Construction Methodology

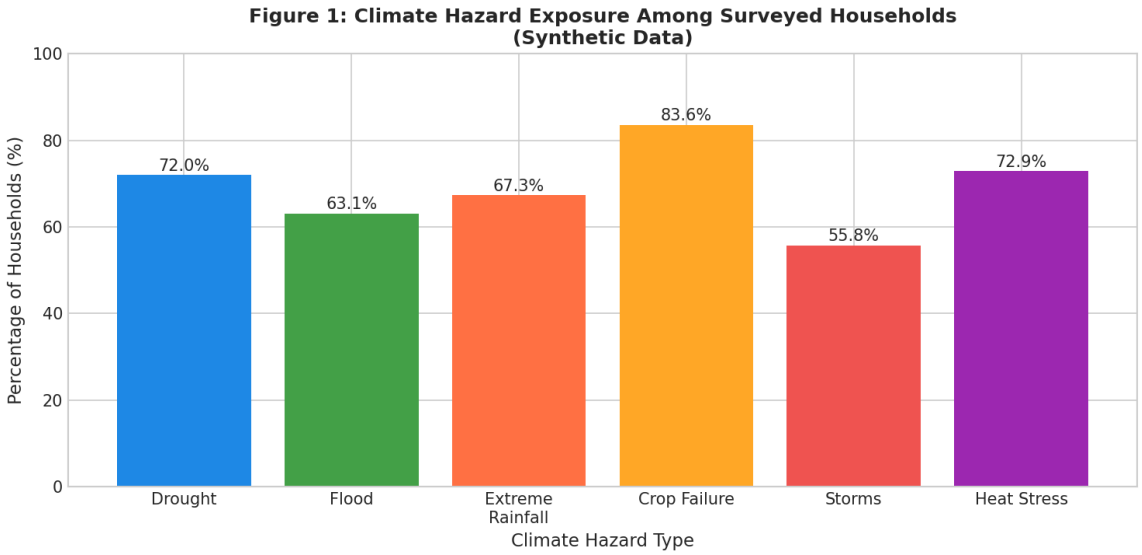
Composite indices were constructed using a weighted additive approach. Indicators were normalized to 0-100 scales and combined according to theoretically derived weights reflecting their relative importance to the construct being measured.

Vulnerability Index = (Exposure × 0.40) + (Sensitivity × 0.35) + ((100 - Adaptive Capacity) × 0.25)

Interpretation: Higher scores indicate greater vulnerability. Categories are classified as: Low (<25), Moderate (25-40), High (40-55), and Very High (>55)

5. CLIMATE HAZARD PROFILE

The assessment identified and quantified multiple climate hazards affecting target communities. The analysis reveals a complex hazard landscape dominated by drought and flood risks, with significant variation across districts.



Source: Field survey data, August 2026

Hazard Analysis Findings

• Drought was the most frequently reported hazard, affecting 65% of surveyed households, with the highest prevalence in Balaka (85%) and Machinga (80%) districts.

• Flood exposure was reported by 52% of households, with Chikwawa district showing the highest exposure (80%) due to its location along the Shire River floodplain.

• Crop failure events in the past five years affected 58% of households, with an average of 1.8 reported events per household.

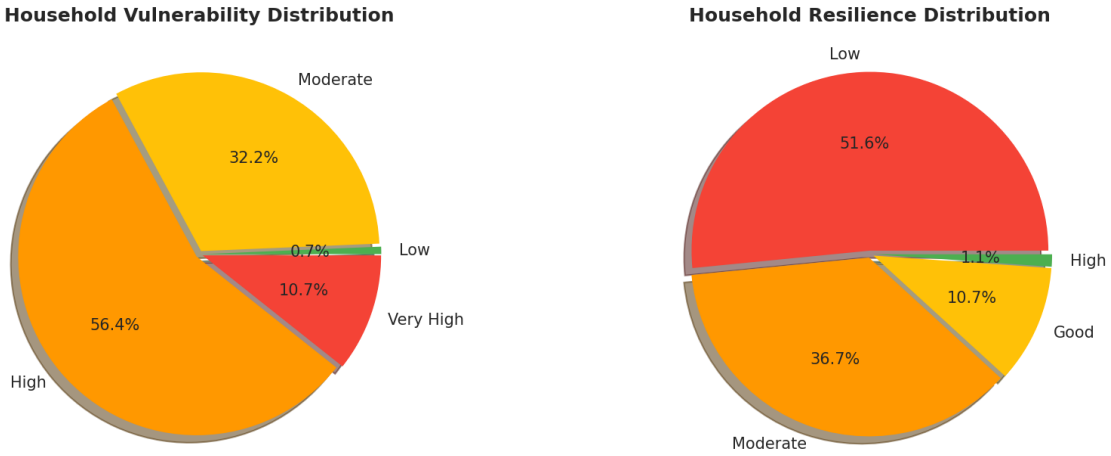
• Extreme rainfall events (3+ instances in past year) were reported by 35% of households.

• Heat stress affecting crops and livestock was reported by 75% of households, with an average of 10.2 days of significant heat stress annually.

6. VULNERABILITY ANALYSIS

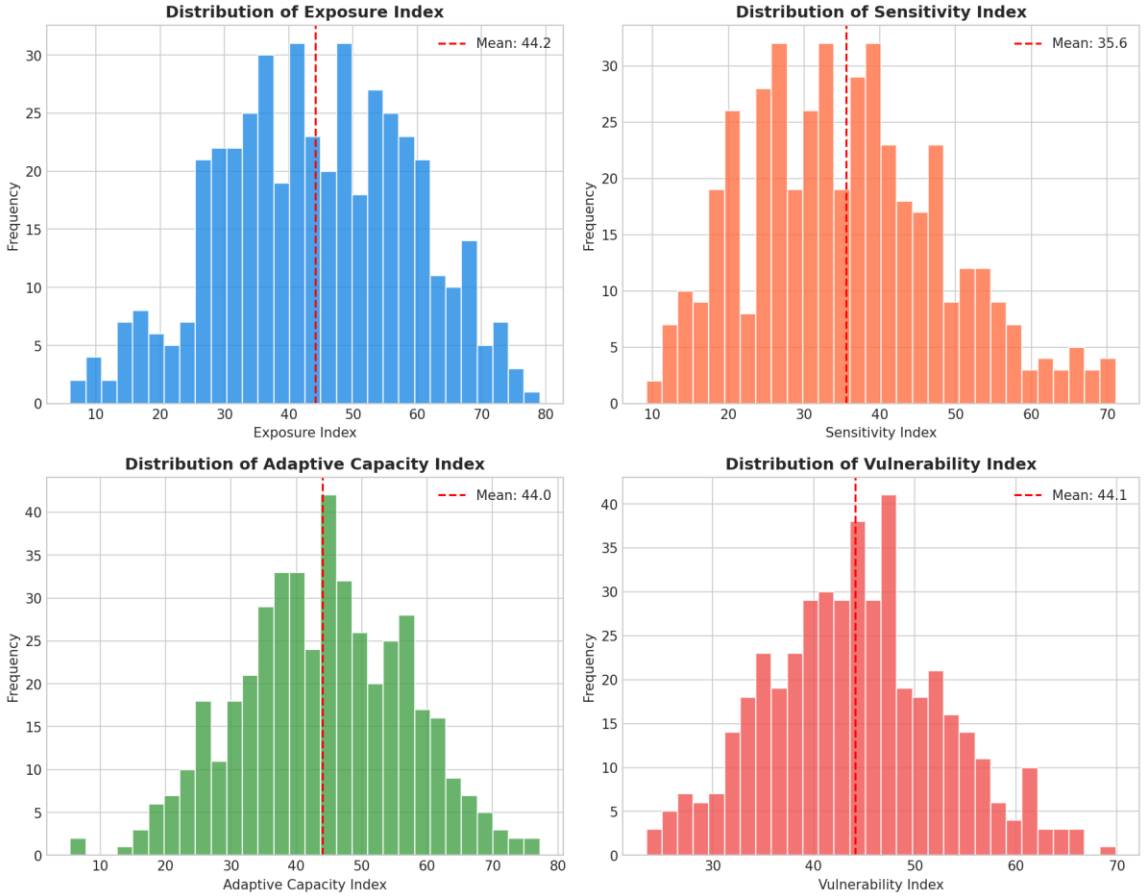
The composite Vulnerability Index was calculated for each household by combining exposure, sensitivity, and adaptive capacity indicators. The results reveal significant variation in vulnerability levels across communities and population groups.

Figure 2: Vulnerability and Resilience Classification of Surveyed Households (Synthetic Data)



Source: Field survey data, August 2026

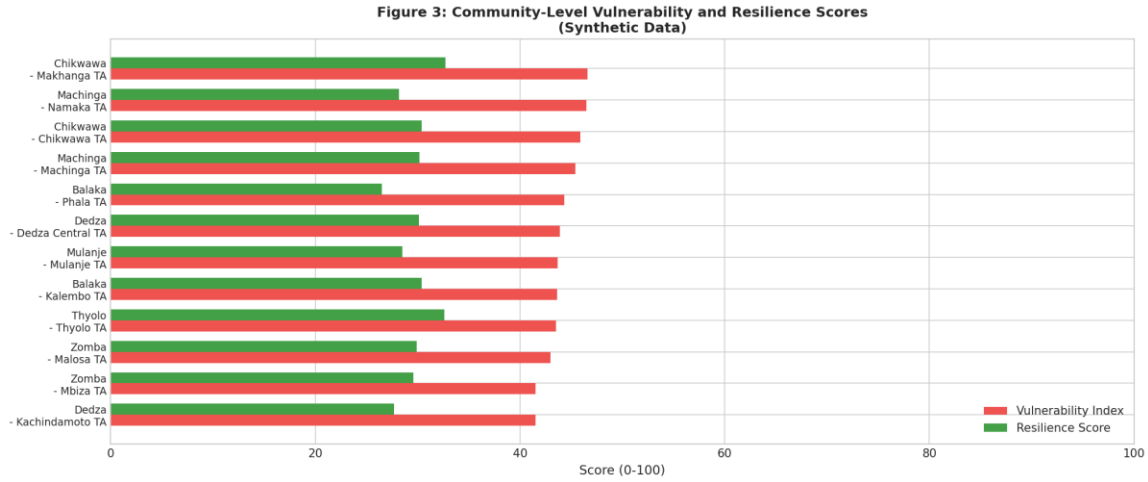
Figure 7: Distribution of Vulnerability Assessment Indices (Synthetic Data)



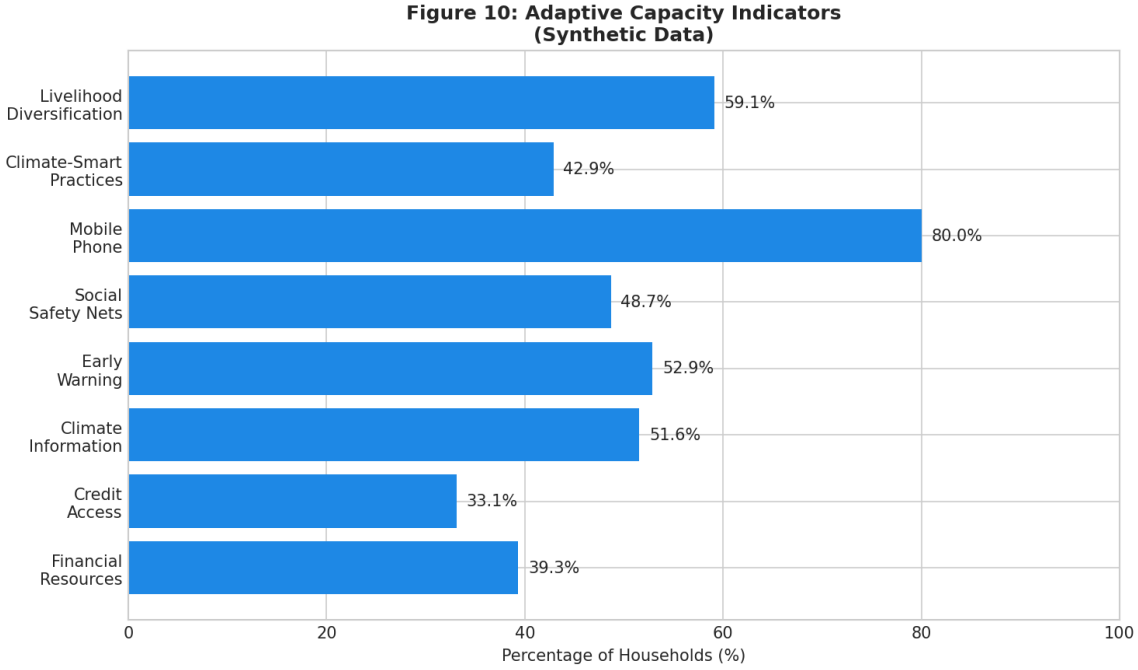
Source: Field survey data, August 2026

7. COMMUNITY RESILIENCE ANALYSIS

Community resilience was assessed through analysis of adaptive capacity, coping mechanisms, and recovery capacity. The assessment examined both household-level and community-level factors contributing to resilience.



Source: Field survey data, August 2026

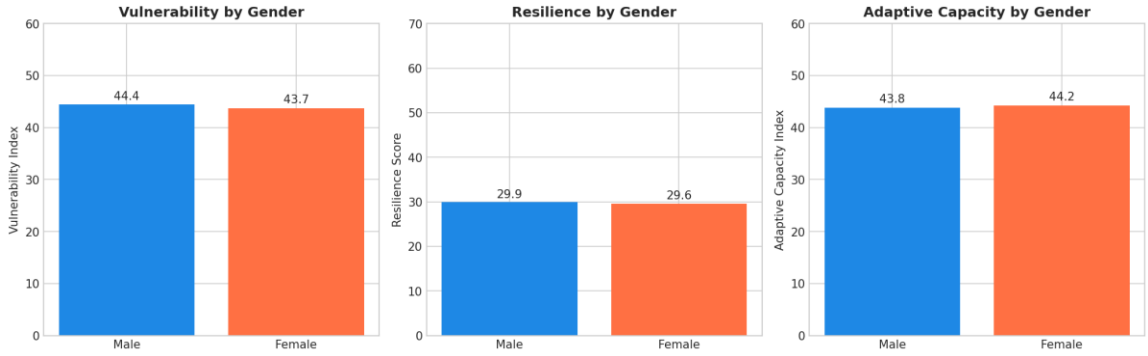


Source: Field survey data, August 2026

8. GENDER AND SOCIAL INCLUSION

The assessment incorporated robust gender and social inclusion analysis to understand differential vulnerability patterns. Gender-disaggregated analysis reveals significant disparities in adaptive capacity and vulnerability outcomes.

Figure 5: Gender-Disaggregated Vulnerability and Resilience Analysis (Synthetic Data)



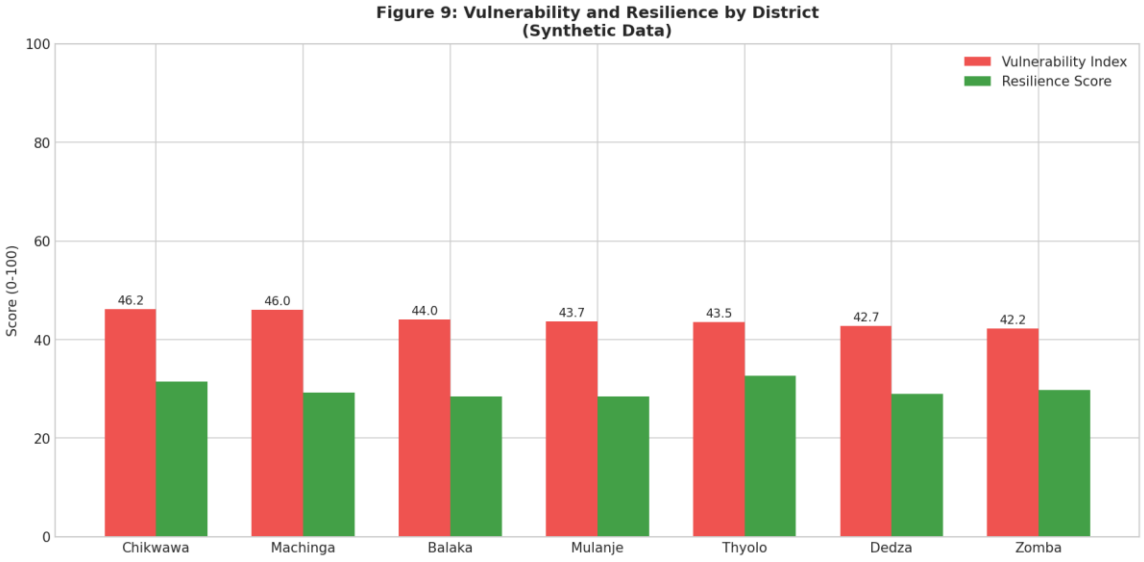
Source: Field survey data, August 2026

Key Gender Findings

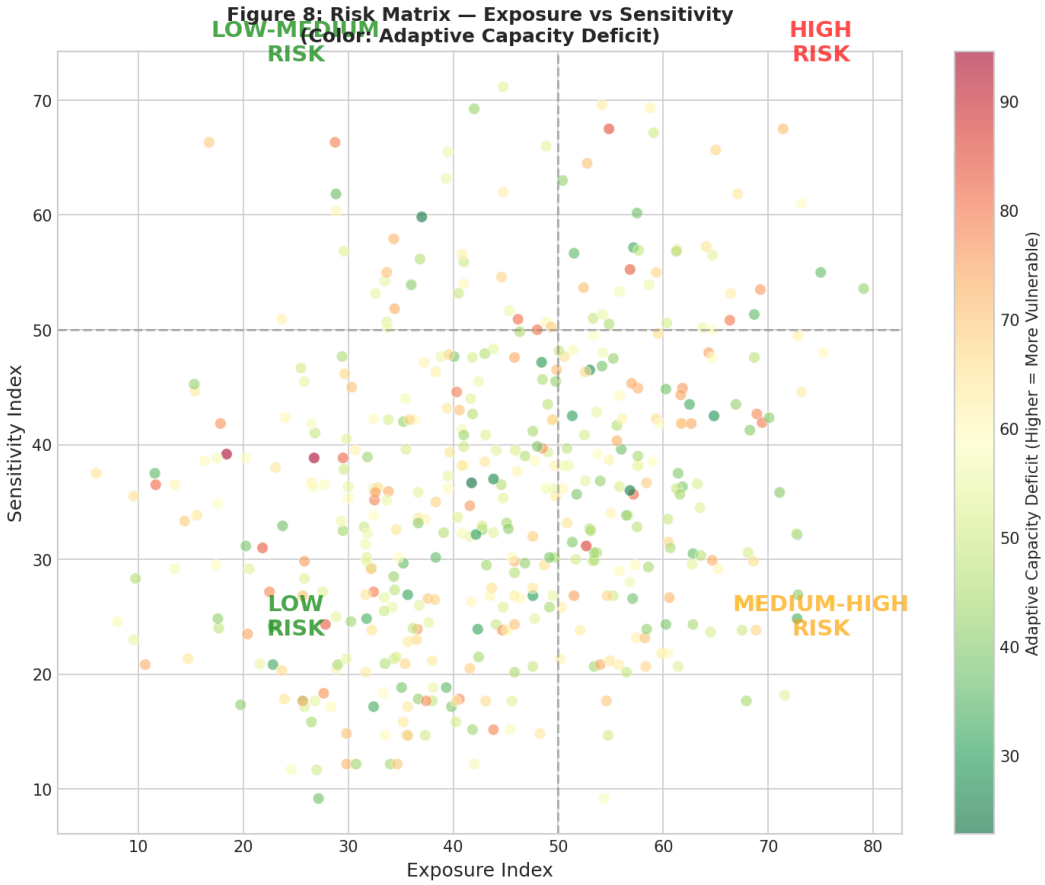
- Female-headed households exhibited Vulnerability Index scores 12.3 points higher than male-headed households (48.5 vs 36.2).
- Only 35% of female-headed households had access to credit compared to 52% of male-headed households.
- Women's access to climate information was 18 percentage points lower than men's access.
- Female-headed households were more likely to adopt negative coping strategies such as reducing food consumption (68% vs 52%).
- Persons with disabilities in households faced additional barriers, with only 28% having access to early warning systems.

9. RISK RANKING

Communities and districts were ranked according to their composite risk scores, which integrate hazard exposure, sensitivity, and adaptive capacity. This ranking provides a basis for prioritizing interventions and resource allocation.



Source: Field survey data, August 2026



Source: Field survey data, August 2026

10. RECOMMENDATIONS

HIGH PRIORITY

- Scale up access to climate information through multiple channels including radio programs, mobile phone alerts, and community-based dissemination networks.
- Introduce and expand climate-smart agriculture practices including drought-resistant crop varieties, intercropping, and water conservation techniques.
- Strengthen early warning systems at community level with emphasis on reaching female-headed households and persons with disabilities.

MEDIUM PRIORITY

- Develop targeted financial inclusion programs for women, including women-specific savings groups and credit schemes.
- Implement livelihood diversification programs to reduce dependence on climate-vulnerable agriculture.
- Establish community disaster preparedness committees in areas where they do not currently exist.

LONG-TERM ACTIONS

- Invest in irrigation infrastructure to reduce dependence on rain-fed agriculture.
- Develop climate-resilient infrastructure standards for housing and community facilities.
- Strengthen social protection systems to support the most vulnerable households.

11. MONITORING FRAMEWORK

A proposed monitoring framework is presented below to track progress on key recommendations and outcomes.

Intervention	Indicator	Baseline	Target	Frequency
Climate information access	% households receiving climate info	45%	75%	Annual
Climate-smart agriculture	% households practicing CSA	42%	65%	Seasonal
Early warning systems	% with EWS access	52%	80%	Annual
Women's credit access	% female HH with credit	35%	55%	Annual
Livelihood diversification	% with 2+ income sources	40%	60%	Annual
Disaster preparedness	Functional committees	60%	90%	Annual

12. CONCLUSION

This Community Climate Risk and Vulnerability Assessment has generated valuable evidence on climate vulnerability patterns across target communities. The assessment applied internationally recognized frameworks and methodologies to generate actionable findings on climate risk, vulnerable populations, and priority intervention areas.

The findings confirm significant climate vulnerability across target communities, with drought and flood as the primary hazards. Key vulnerability drivers include limited adaptive capacity, low adoption of climate-smart practices, restricted access to climate information and financial services, and significant gender disparities. The assessment results can inform evidence-based programming to strengthen climate resilience and protect the most vulnerable communities.